

Session Info

Ivan Jacob Agaloos Pesigan

Session

```
sessionInfo()

#> R version 4.5.3 (2026-03-11)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS: /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblas-p0.3.26.so; LAPACK version 3.12.0
#>
#> locale:
#>  [1] LC_CTYPE=en_US.UTF-8      LC_NUMERIC=C
#>  [3] LC_TIME=en_US.UTF-8      LC_COLLATE=en_US.UTF-8
#>  [5] LC_MONETARY=en_US.UTF-8  LC_MESSAGES=en_US.UTF-8
#>  [7] LC_PAPER=en_US.UTF-8     LC_NAME=C
#>  [9] LC_ADDRESS=C             LC_TELEPHONE=C
#> [11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: Etc/UTC
```

```

#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats      graphics  grDevices  utils      datasets  methods   base
#>
#> other attached packages:
#> [1] rProject_0.0.25
#>
#> loaded via a namespace (and not attached):
#> [1] backports_1.5.0  R6_2.6.1          lubridate_1.9.5  xfun_0.57
#> [5] magrittr_2.0.4   glue_1.8.0        stringr_1.6.0    knitr_1.51
#> [9] timechange_0.4.0 generics_0.1.4    lifecycle_1.0.5  xml2_1.5.2
#> [13] cli_3.6.5.9000  bibtex_0.5.2      compiler_4.5.3   plyr_1.8.9
#> [17] highr_0.12       rprojroot_2.1.1   httr_1.4.8       tools_4.5.3
#> [21] evaluate_1.0.5   Rcpp_1.1.1        RefManageR_1.4.0 otel_0.2.0
#> [25] rlang_1.1.7      jsonlite_2.0.0    stringi_1.8.7

```

Packages

```

unname(installed.packages()[, 1])

#> [1] "metaDyn"      "abind"         "Amelia"
#> [4] "arm"          "arrow"         "ash"
#> [7] "AsioHeaders"  "askpass"       "assertthat"
#> [10] "backports"    "base64enc"     "betaDelta"
#> [13] "betaMC"       "betaNB"        "betaSandwich"
#> [16] "BH"           "bibtex"        "BiocManager"

```

```

#> [19] "BiocVersion"      "bit"              "bit64"
#> [22] "bitops"           "blob"             "bootStateSpace"
#> [25] "brew"             "brio"             "broom"
#> [28] "bslib"            "cachem"           "callr"
#> [31] "car"              "caracas"          "carData"
#> [34] "cellranger"       "cffr"             "checkmate"
#> [37] "cli"              "clipr"            "clock"
#> [40] "clusterGeneration" "coda"             "cOde"
#> [43] "collections"      "colorspace"       "commonmark"
#> [46] "conflicted"       "corpcor"          "covr"
#> [49] "cowplot"          "cpp11"            "crayon"
#> [52] "credentials"      "crosstalk"        "cTMed"
#> [55] "ctsem"            "curl"             "data.table"
#> [58] "DBI"              "dbplyr"           "Deriv"
#> [61] "desc"             "deSolve"          "devtools"
#> [64] "diagram"          "dials"            "DiceDesign"
#> [67] "diffobj"          "digest"           "distributional"
#> [70] "distro"           "doBy"             "docopt"
#> [73] "downlit"          "dplyr"            "DT"
#> [76] "dtplyr"           "duckdb"           "dynr"
#> [79] "dynUtils"         "ellipse"          "ellipsis"
#> [82] "evaluate"         "expm"             "fansI"
#> [85] "farver"           "fastDummies"      "fastmap"
#> [88] "fclust"           "fda"              "fdrtool"
#> [91] "fds"             "fitVARMxID"       "FNN"
#> [94] "fontawesome"      "forcats"          "foreach"
#> [97] "forecast"         "Formula"          "fracdiff"

```

```

#> [100] "fs"           "fst"           "fstcore"
#> [103] "furrr"        "future"        "future.apply"
#> [106] "gargle"       "generics"      "gert"
#> [109] "ggplot2"      "ggrepel"       "gh"
#> [112] "gitcreds"     "glasso"        "glmnet"
#> [115] "globals"     "glue"          "googledrive"
#> [118] "googlesheets4" "gower"         "GPArotation"
#> [121] "GPfit"        "graphicalVAR"  "gridExtra"
#> [124] "gsubfn"       "gtable"        "gtools"
#> [127] "hardhat"      "haven"         "hdcrcde"
#> [130] "here"         "highr"         "Hmisc"
#> [133] "hms"          "htmlTable"     "htmltools"
#> [136] "htmlwidgets" "httpgd"        "httpuv"
#> [139] "httr"         "httr2"         "ids"
#> [142] "ifaTools"     "igraph"        "infer"
#> [145] "ini"          "inline"        "ipred"
#> [148] "isoband"      "iterators"     "jomo"
#> [151] "jpeg"         "jquerylib"     "jsonlite"
#> [154] "jsonvalidate" "kernlab"       "knitr"
#> [157] "ks"           "kutils"        "labeling"
#> [160] "Lahman"       "languageserver" "later"
#> [163] "latex2exp"    "lava"          "lavaan"
#> [166] "lazyeval"     "lhs"           "lifecycle"
#> [169] "lintr"        "lisrelToR"     "listenv"
#> [172] "litedown"     "littler"       "lme4"
#> [175] "lmtest"       "locfit"        "longMI"
#> [178] "loo"          "lubridate"     "magick"

```

```

#> [181] "magrittr"      "markdown"      "MatrixModels"
#> [184] "matrixStats"  "mclust"        "memoise"
#> [187] "metaDyn"      "metaSEM"       "mi"
#> [190] "mice"         "microbenchmark" "mime"
#> [193] "miniUI"       "minqa"         "mitml"
#> [196] "mize"         "mlVAR"         "mnormt"
#> [199] "modeldata"    "modelenv"      "modelr"
#> [202] "MplusAutomation" "multicool"    "mvtnorm"
#> [205] "nloptr"       "numDeriv"      "nycflights13"
#> [208] "OpenMx"       "openssl"       "openxlsx"
#> [211] "ordinal"      "otel"          "pak"
#> [214] "pan"          "pander"        "parallelly"
#> [217] "parsnip"      "patchwork"     "pbapply"
#> [220] "pbivnorm"     "pbkrtest"      "pcaPP"
#> [223] "pdftools"     "pillar"        "pkgbuild"
#> [226] "pkgconfig"    "pkgdown"       "pkgload"
#> [229] "plyr"         "png"           "posterior"
#> [232] "pracma"       "praise"        "prettyunits"
#> [235] "printr"       "processx"      "prodlim"
#> [238] "profvis"      "progress"      "progressr"
#> [241] "promises"     "proto"         "ps"
#> [244] "psych"        "purrr"         "qgraph"
#> [247] "qpdf"         "quadprog"      "quantreg"
#> [250] "quarto"       "QuickJSR"      "R.cache"
#> [253] "R.methodsS3"  "R.oo"          "R.utils"
#> [256] "R2jags"       "R2WinBUGS"     "R6"
#> [259] "ragg"         "rainbow"       "rappdirs"

```

```

#> [262] "rbibutils"      "rcmdcheck"      "RColorBrewer"
#> [265] "Rcpp"           "RcppArmadillo"  "RcppEigen"
#> [268] "RcppGSL"        "RcppParallel"   "RcppTOML"
#> [271] "RCurl"          "Rdpack"         "readr"
#> [274] "readxl"         "recipes"        "RefManager"
#> [277] "reformulas"     "rematch"        "rematch2"
#> [280] "remotes"        "reprex"         "reshape2"
#> [283] "reticulate"     "rex"            "rhdf5"
#> [286] "rhdf5filters"   "Rhdf5lib"       "rhub"
#> [289] "rjags"          "rlang"          "RMariaDB"
#> [292] "rmarkdown"     "rockchalk"      "roxygen2"
#> [295] "rpf"           "RPostgres"      "rProject"
#> [298] "rprojroot"     "rsample"        "RSQLite"
#> [301] "rstan"         "rstantools"     "rstudioapi"
#> [304] "RUnit"         "rversions"      "rvest"
#> [307] "Ryacas"        "S7"            "sass"
#> [310] "scales"        "selectr"        "sem"
#> [313] "semblbci"      "semmcci"        "semPlot"
#> [316] "sessioninfo"   "sfd"           "shape"
#> [319] "shiny"         "simStateSpace"  "slider"
#> [322] "snow"          "snowfall"       "sourcetools"
#> [325] "SparseM"       "sparsevctrs"    "SQUAREM"
#> [328] "StanHeaders"   "stringi"        "stringr"
#> [331] "styler"        "symSEM"         "sys"
#> [334] "systemfonts"   "tailor"         "tensorA"
#> [337] "testthat"      "texreg"         "textshaping"
#> [340] "tibble"        "tidymodels"     "tidyr"

```

```

#> [343] "tidyselect"      "tidyverse"      "timechange"
#> [346] "timeDate"       "tinytex"        "tune"
#> [349] "tzdb"           "ucminf"         "unigd"
#> [352] "urca"           "urlchecker"     "usethis"
#> [355] "utf8"           "uuid"           "V8"
#> [358] "vctrs"          "viridisLite"    "vroom"
#> [361] "waldo"          "warp"           "whisker"
#> [364] "whoami"         "withr"          "workflows"
#> [367] "workflowsets"   "xfun"           "XML"
#> [370] "xml2"           "xmlparsedata"   "xopen"
#> [373] "xtable"         "yaml"           "yardstick"
#> [376] "zip"            "zoo"            "base"
#> [379] "boot"           "class"          "cluster"
#> [382] "codetools"      "compiler"       "datasets"
#> [385] "foreign"        "graphics"       "grDevices"
#> [388] "grid"           "KernSmooth"     "lattice"
#> [391] "MASS"           "Matrix"         "methods"
#> [394] "mgcv"           "nlme"           "nnet"
#> [397] "parallel"       "rpart"          "spatial"
#> [400] "splines"        "stats"          "stats4"
#> [403] "survival"       "tcltk"          "tools"
#> [406] "utils"

```

References

R Core Team. (2026). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. Vienna, Austria. <https://www.R-project.org/>